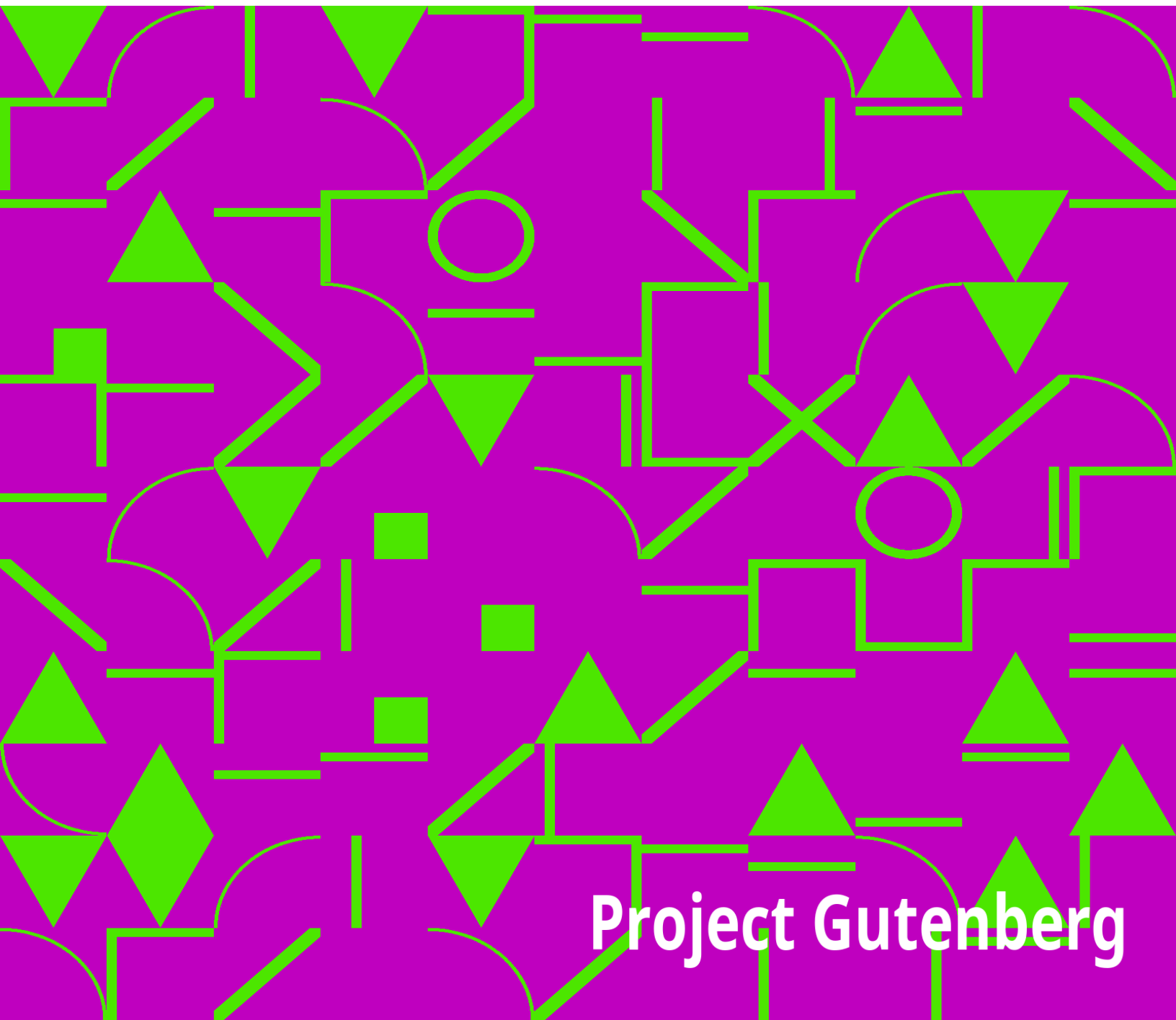


Subspeciation in the Meadow Mouse, *Microtus pennsylvanicus*, in Wyoming, Colorado, and Adjacent Areas

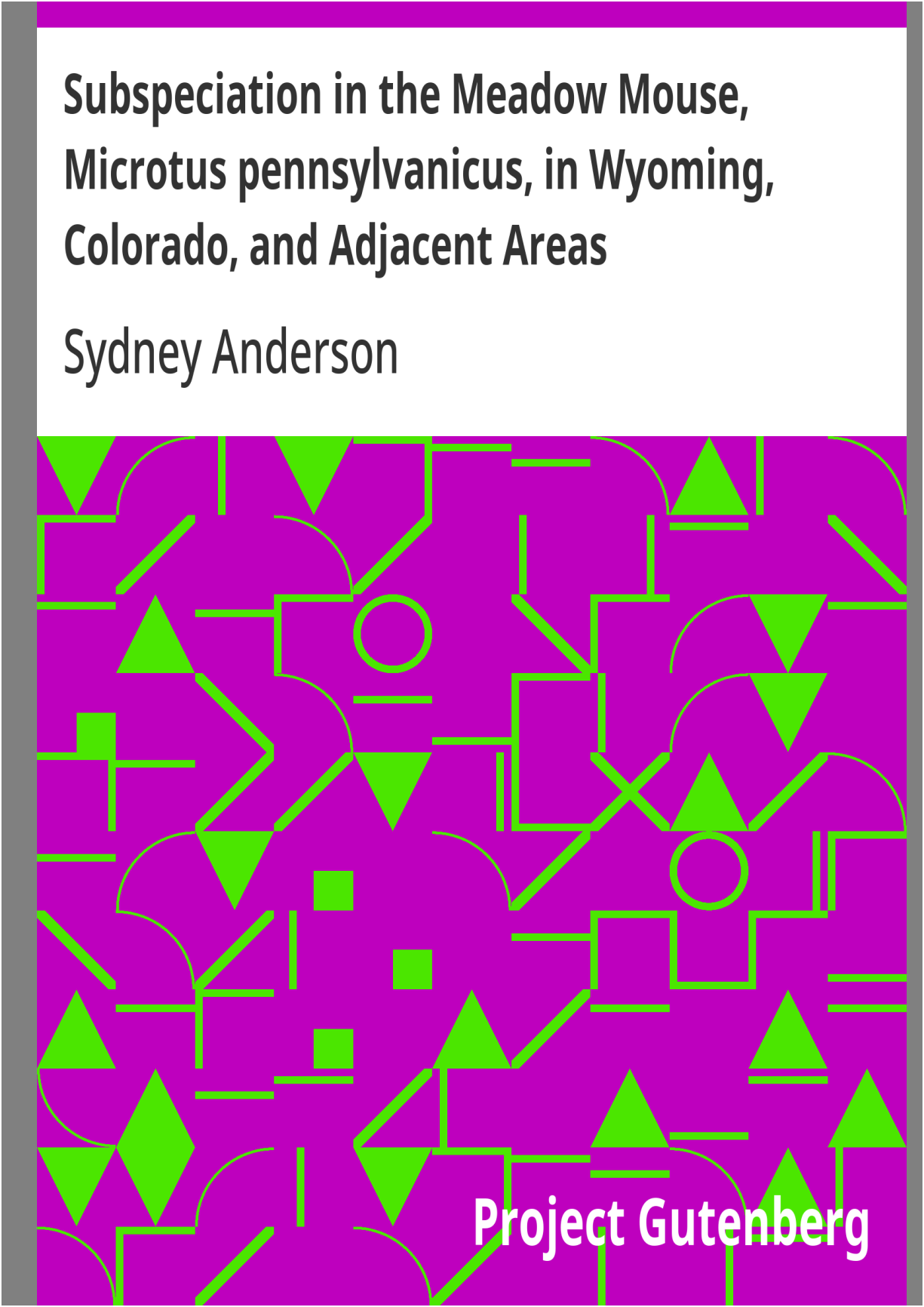
Sydney Anderson

An abstract geometric pattern consisting of various shapes like triangles, squares, circles, and lines in shades of blue, green, and yellow on a black background.

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Sydney Anderson



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*** START OF THE PROJECT GUTENBERG EBOOK SUBSPECIATION IN THE
MEADOW MOUSE, *MICROTUS PENNSYLVANICUS*, IN WYOMING, COLORADO,
AND ADJACENT AREAS ***

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BY

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**UNIVERSITY OF KANSAS
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INTRODUCTION

In the region including Wyoming and Colorado, *Microtus pennsylvanicus* has been divided into two subspecies: the pale *M. p. insperatus* (J. A. Allen) inhabits the Black Hills of the northeasternmost part of Wyoming; the dark *M. p. modestus* (Baird) inhabits extensive areas in both Wyoming and Colorado. Initial examination of *Microtus pennsylvanicus* revealed that specimens from the Big Horn Mountains of north-central Wyoming (within the range of *modestus* as mapped by Hall and Cockrum 1952:407), in color at least, resemble the subspecies *insperatus* more than they do *modestus*, and that specimens from southwestern Wyoming are notably dark. Durrant (1952:363) noted that specimens from Utah are dark, and Davis (1939:315) did the same for specimens from near Pocatello, Idaho. It seemed, therefore, that dark color might characterize populations of a wide geographic region and distinguish them from *modestus* named from southern Colorado. Also, there seemed to be a hiatus of at least 180 miles between the ranges of *modestus* in northern Colorado and *modestus* in eastern Wyoming, and an even greater distance separating populations of *modestus* in northern Colorado from those in western Wyoming. *Microtus pennsylvanicus* has not been taken in central or southeastern Wyoming despite extensive collecting there, which yielded numerous records of other kinds of *Microtus* (*M. longicaudus*, *M. montanus*, and *M. ochrogaster*). Subsequent study revealed a pattern of geographic variation within the range now ascribed to *modestus* which, in my opinion, can be described best by the recognition of three new subspecies.

MATERIALS, METHODS, AND ACKNOWLEDGMENTS

To study geographic variation in color a method was devised as follows: A single skin (KU 42407, from 1½ miles east of Buckhorn in Weston County, Wyoming) was selected as a representative of the paler mice and arbitrarily given the number 2. A single skin (KU 17491, from 3 miles east of Moran in Teton County, Wyoming) was selected as a representative of the darker mice from the western part of Wyoming and arbitrarily given the number 4. These mice were selected so that they were respectively paler and darker than the estimated average of the total variation within the populations to be studied, but the two mice were not at the extremes of paleness and darkness. Comparisons were based on visual inspection of the dorsal pelage as a whole. Skins were compared with these two mice and given whole numbers from one to five. If paler than the standard for 2, the skin was numbered one; if not distinguishably paler or darker, it was given the number two; if intermediate in color to the standards for 2 and 4 and not definitely more nearly referable to one than the other, it was given the number three; if it resembled the standard for 4, it was numbered four; and if darker, it was given the number five. In this manner skins from a given locality could be evaluated one by one and the results plotted, averaged, and treated statistically. On Figure 1 the average values for color of 32 series are mapped to show the geographic variation of color. The following series of adults are the basis for Figure 1 (abbreviations for collections other than at the University of Kansas are included in parentheses): Each locality is followed by the month (or months) of capture, the number of specimens, and the average value for color.

Montana: Glacier County, August, 6, 1.8; Hill and Chouteau counties combined (Mich), July, 24, 1.5; Malta, Philips County, August, 14, 1.5; Sheridan County, August, 6, 1.5; Fergus County (USBS), August, 5, 2.4; Ravalli County (KU and USBS), August, 12, 2.8; Silver Bow County, August, 7, 3.0; Sweet Grass County (Mich), June and July, 7, 2.7; Park County, August, 10, 2.6. *Idaho*: Pocatello and vicinity, November and December, 5, 3.4. *Wyoming*: Park County, August, 6, 2.8; Sheridan County, September, 9, 1.2; Johnson County, August, 12, 1.5; Campbell and Crook counties, July, 11, 1.4; Weston County, July, 7, 1.6; Teton County, September, 8, 3.4; Teton County (Mich), June, 17, 3.1; Afton and vicinity, Lincoln County, July, 10, 4.2; Sage, Lincoln County, July, 5, 5.0. *South Dakota*: Pennington County (Chi), June, 14, 2.1; Pennington County (Mich), December and January, 8, 1.1; Walworth County, July, 4, 3.7; Buffalo County, July, 6, 3.2. *Colorado*: Loveland and vicinity, Larimer County (KU and USBS), July, 13, 2.8; Boulder County (Chi), September, 34, 2.6; Park County (Denv), March, 8, 1.9; Colorado Springs (ERW), March, April, and May, 5, 2.8; Saguache County (USBS), August, 46, 3.0; Conejos County, June, 4, 3.0; Wray, Yuma County (USBS), 3, 4.7. *Nebraska*: Dundy County, August and November, 14, 4.6. *New Mexico*: Colfax County, June, 8, 3.2. Variation in color is discussed in the accounts of the subspecies concerned.

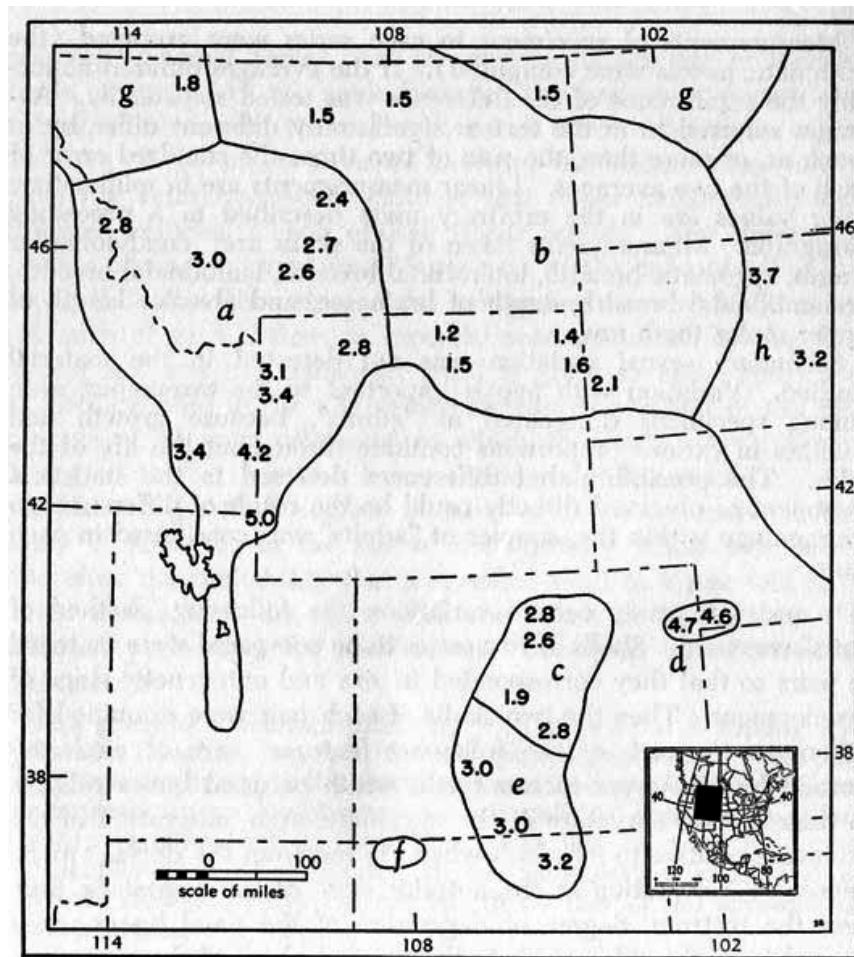


FIGURE 1. Geographic variation in color in *Microtus pennsylvanicus* in the Rocky Mountains. Paler colors are represented by smaller numbers. Numbers are derived from the series of specimens listed in the text by the method described there. The subspecies that occur in the region studied are as follows:

- | | |
|----------------------------|--------------------------------|
| a. <i>M. p. pullatus</i> | e. <i>M. p. modestus</i> |
| b. <i>M. p. insperatus</i> | f. <i>M. p. aztecus</i> |
| c. <i>M. p. uligocola</i> | g. <i>M. p. drummondi</i> |
| d. <i>M. p. finitus</i> | h. <i>M. p. pennsylvanicus</i> |

For each of the series listed in Table 1 all adult mice having skulls that measured more than 24.0 mm. in condylobasilar length were studied. Total length, length of tail, and length of hind foot were taken from the collector's field labels. The measurements of the skulls listed below were taken by means of dial calipers reading to one-tenth of a millimeter, and in the same fashion as described previously (Anderson, 1954:492). Measurements of specimens in each series were averaged (the arithmetic means were computed). If the averages differed noticeably the significance of the difference was tested statistically. Averages referred to in the text as significantly different differ by as much as, or more than, the sum of two times the standard error of each of the two averages. Linear measurements are in millimeters; color values are in the arbitrary units described in a preceding paragraph. Measurements taken of the skulls are: condylobasilar length, zygomatic breadth, interorbital breadth, lambdoidal breadth, prelamdoidal breadth, depth of braincase, and alveolar length of upper molar tooth-row.

Secondary sexual variation was not detected in the material studied. Variation with age is important to the taxonomist even among specimens designated as "adults", because growth and changes in various proportions continue throughout the life of the mice. The possibility that differences detected in the statistical treatment or observed directly could be the result of differences in average age within the samples of "adults" was considered in each case.

In order to study certain variations, the following "method of pairs" was used. Skulls of two series to be compared were matched in pairs so that they corresponded in size and ontogenetic stage of development. Then the two skulls of each pair were examined for differences in each of the following features: size of circle inscribed by the upper incisor teeth, width of nasal bones relative to their length, curvature of the zygomatic arch, elongation of the braincase relative to its width when viewed from the dorsal aspect, degree of indentation in the anterior edge of the zygomatic arch near the rostrum, degree of depression of the nasal bones when viewed from the side, width in the vertical plane of the zygomatic arch at the suture between the maxillary and jugal bones, length relative to width of the prominent fenestra in the posterodorsal part of the squamosal bone, size of the meatus of the auditory canal, distance between the internal margin of an occipital condyle at its posteriormost point and the tip of the paraoccipital process of the same side of the skull, size of the foramen magnum, vertical height

of the supraoccipital bone from the dorsalmost point on the margin of the foramen magnum to the midpoint of the lambdoidal crest, constriction posteriorly or narrowness of the incisive foramen relative to its length, distance across the premaxillary bone from the anteriormost point of the incisive foramen to the posteriormost point of the margin of the alveolus of the upper incisor, area of the maxillary septum (Howell 1926:112, or "zygomatic plate" of Ellerman 1941:1), acuminateness of the anterior border of the palatine opening (internal nares), size of auditory bullae, size of foramen ovale, acuteness of the angle between the basioccipital and basisphenoidal bones at the suture between them (degree to which the area of the suture is raised between the bullae when viewed from the ventral aspect), width of first upper molar tooth, least distance between alveoli of first upper molars. Any differential feature present in more than 75 per cent of the pairs of animals is reported in the discussion of the subspecies concerned. The significance of each difference reported was calculated by the Chi-square test and the confidence limit is given in each case. The probability used in the Chi-square formula is one-half of the percentage of all pairs compared in which the skulls were different in regard to the character being considered. For example, in 68 per cent of the total number of pairs of skulls compared in this study a difference in the size of the auditory bullae was noted. Therefore the probability that a specified skull of a pair will have larger bullae than the other skull was taken as 34 per cent. A different probability for each feature compared was derived in like manner.

This study is concerned primarily with mice from Wyoming and Colorado; I realize, however, that the physiographic and ecological conditions important to the distribution and subspeciation of *Microtus pennsylvanicus* do not correspond to political boundaries. Geographic variation within these two states can be seen in proper perspective only when related to the neighboring areas and to previous studies. I have attempted to do this in the accounts of the subspecies.

Approximately five months in the field in Wyoming and Colorado in the summers of 1950, 1951, 1952, and 1953 gave me a familiarity with the region that has helped to clarify the pattern of distribution. My study was based, in addition, on 762 specimens that are listed under "specimens examined" in the accounts of subspecies, and on comparative material from other states. Most of these specimens are skins with skulls but some are skins only and others are skulls only. Some localities are represented by too few adult individuals to permit significant comparisons. Owing to damaged skulls, certain measurements of some specimens were omitted from the calculations. If it seemed that the damaged skull was exceptionally large or small or a deviant in any other regard it was not used, in order not to bias the computed averages, which might be used in comparing proportions of the skulls. In the lists of specimens examined, localities that are omitted from Figure 2 because overlapping or undue crowding of the symbols would have resulted are *italicized*.

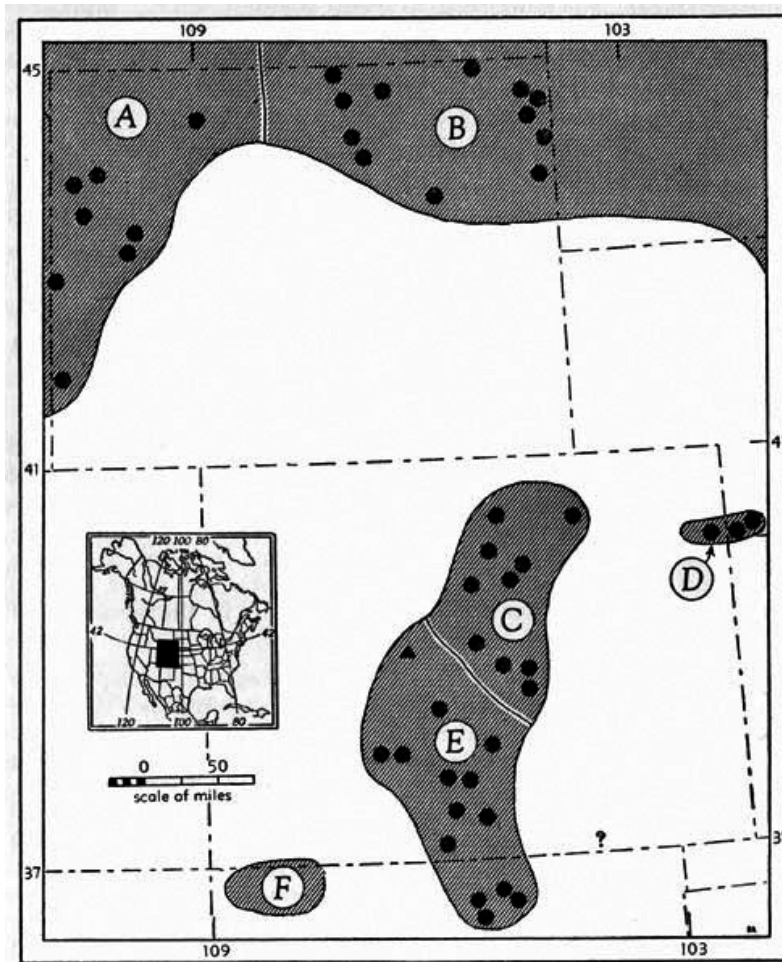


FIGURE 2. Distribution of the subspecies of *Microtus pennsylvanicus* in Wyoming and Colorado. Solid dots represent localities from which specimens have been examined, and triangles represent localities reported in the literature from which I have not examined specimens. The question mark in southern Colorado denotes a questionable record discussed in the text.

- | | |
|----------------------------|--------------------------|
| A. <i>M. p. pullatus</i> | D. <i>M. p. finitus</i> |
| B. <i>M. p. insperatus</i> | E. <i>M. p. modestus</i> |
| C. <i>M. p. uligocola</i> | F. <i>M. p. aztecus</i> |

I am grateful to Professor E. Raymond Hall for critical reading of the manuscript and helpful suggestions, to Dr. Rollin H. Baker and various of my fellow students at the Museum of Natural History for stimulating comments pertinent to the problems involved in this study, to my wife, Justine Anderson, for assistance in the preparation of the manuscript, to numerous members of field parties from the Museum of Natural History, who collected much of the material studied, and to the curators and other persons, at the museums listed below, who courteously made specimens available for study. The field work of the Museum of Natural

History was assisted by the Kansas University Endowment Association. A National Science Foundation Fellowship made it possible for me to visit the museums listed below. An honorarium awarded by the American Society of Mammalogists enabled me to present this paper at the 34th Annual Meeting of the Society, in June of 1954. Unless otherwise indicated, specimens are in the University of Kansas Museum of Natural History. Specimens in other museums are designated beyond as follows: American Museum of Natural History (AMNH); Museum of Zoology, University of Michigan (Mich); Chicago Natural History Museum (Chi); United States National Museum (USNM); Biological Surveys Collection (USBS); Museum of Vertebrate Zoology, Berkeley, California (MVZ); Colorado Museum of Natural History, Denver (Denv); E. R. Warren collection, Colorado College, Colorado Springs (ERW); University of Colorado Museum, Boulder (UC).

ACCOUNTS OF SUBSPECIES

***Microtus pennsylvanicus modestus* (Baird)**

Arvicola modesta Baird, Repts. Expl. and Surv...., pt. 1, Mammals, p. 535, July 14, 1858.

Microtus pennsylvanicus modestus, Bailey, N. Amer. Fauna, 17:20, June 6, 1900.

Type.—Immature specimen (sex not specified), skin and skull, number of skin 594, number of skull 1717, deposited in the collections of the Smithsonian Institution, obtained by F. Kreutzfeldt, Sawatch Pass, Rocky Mountains [=Saguache Pass or Cochetopa Pass, Saguache County, Colorado], exact date unknown. I have not examined the type specimen.

Range.—Northern New Mexico, and southern Colorado (see list of specimens and Fig. 2).

Comparisons.—For comparison with the subspecies newly described below from northern Colorado see the account of that subspecies. The subspecies *M. p. aztecus* has been compared with *M. p. modestus* by Hall and Cockrum (1952:308) who reduced *aztecus* to subspecific rank. Although *aztecus* is separated by approximately 100 miles from *modestus*, and although no proof of intergradation is available, my studies of variation in this species lead me to agree with Hall and Cockrum that "the morphological differences between the two kinds of animals are of the degree and kind that separate subspecies, rather than species." A more adequate series of adults of *aztecus* is needed to clarify even the subspecific differences between *aztecus* and *modestus*.

Measurements.—Averages, extremes, and standard deviations of a number of series are included in Table 1 in order to facilitate comparisons between different subspecies.

Remarks.—The dividing line between *M. p. modestus* and the subspecies to the north on Figure 2 is drawn somewhat arbitrarily because few specimens are available from this area.

Actual intergradation, in the form of a geographically intermediate population also morphologically intermediate between these two subspecies, is lacking. However, in most populations of both subspecies some individuals are intermediate between the two subspecies or even more like the other subspecies than the one to which they are referred. Warren (1942:226) states that *modestus* has been recorded from Lake County, although no reference to a specimen is given. Bailey (1900:21) cites Twin Lakes, in Lake County. That county is near the dividing line as I have drawn it, and therefore specimens from Lake County would be of special interest. An isolated colony of *modestus* occurs at San Rafael in Valencia County, New Mexico (Bailey, 1932:201). A hiatus of approximately 150 miles separates that colony from the southernmost locality shown in Figure 2. A single specimen, the skin of an immature *Microtus* without skull, from Trinchera, Colorado, taken by L. R. Hersey in 1912, is in the Colorado Museum of Natural History. No species of Microtine has been recorded from within 50 miles of this locality. The specimen is seemingly more like *M. pennsylvanicus* than any other species of *Microtus*. This locality is represented in Figure 2 by a question mark. Ecologically *M. montanus fuscus* Hall and *M. longicaudus mordax* (Merriam) in the same region occupied by *modestus* seem to be more montane than *modestus*. It favors lush grass on the wet floors of alluvial valleys, and also irrigated areas such as that at Manassa.

Specimens examined.—Total 130. *Colorado*: CHAFFEE Co.: Salida, 3 (ERW). SAGUACHE Co.: Monshower Meadows, 27 mi. NW Saguache [=3 mi. E Cochetopa Pass], 8 (USBS); Tevebaugh's Ranch, 20 mi. W Saguache, 46 (USBS, there are additional specimens not examined in detail by me); Cochetopa Pass, 33 mi. W Saguache, 1; 3 mi. N, 16 mi. W Saguache, 8500 ft., 5; 5 mi. NW Hooper, 1 (AMNH); Medano Ranch, 15 mi. NE Mosca, 2 (1 ERW, 1 USBS). CUSTER Co.: Westcliffe, 7800 ft., 1 (ERW). ALAMOSA Co.: Hooper, 10 (2 AMNH, 8 Denv); Mosca, 3 (ERW). CONEJOS Co.: 1½ mi. E Manassa, 11. COSTILLA Co.: Alamosa, 3 (Mich); 2 mi. S Blanca, 7800 ft., 6 (MVZ). *New Mexico*: TAOS Co.: Arroyo Hondo, 7600 ft., 6 (USBS); Taos, Pueblo, 1 (USNM). COLFAX Co.: 1 mi. S, 2 mi. E Eagle Nest, 8100 ft., 21; Taos Mountains, east slope, 8800 ft., 1 (USBS); Coyote Creek, 1 (USBS).

***Microtus pennsylvanicus uligicola* new subspecies**

Type.—Adult male, skin and skull, number 26898, University of Kansas, Museum of Natural History, obtained by James O. Lounquist, original number 349, 6 miles west and ½ mile south of Loveland, 5200 ft., Larimer Co., Colorado, on July 26, 1948.

Range.—Northern Colorado. See Figure 2 and list of specimens examined.

Diagnosis.—Entire animal and skull large; color average for the species, neither extremely pale or dark in summer pelage; molar tooth-row long; nasals narrow; maxillary septum large; first upper molar wide; anterior margin of zygomatic arch above infraorbital foramen not deeply indented; fenestrae in posterodorsal parts of squamosal bones relatively long; braincase not elongate; auditory bullae and meatus large.

Comparisons.—From *M. p. modestus*, *M. p. uligocola* differs as follows: averages paler; prelabial breadth and alveolar length of molar tooth-row significantly greater. Six pairs of skulls were compared. Of the features listed above under the "method of pairs" only two features differed in more than 75 per cent of the pairs; in 5 of 6 pairs *uligocola* had a less distinctly indented anterior margin of the zygomatic arch (Confidence Limit .95) and a more elongate posterodorsal squamosal fenestra (C. L. .85). Seven pairs of skulls from Boulder, Colorado, representing *uligocola* and from Colfax County, New Mexico, representing *modestus* differed in more than 75 per cent of the pairs in three features. Only one of these differences, the elongation of the posterodorsal squamosal fenestra, was the same as a difference noted above between topotypes of *uligocola* and *modestus*. A comparison of ten pairs of skulls of *uligocola* from Boulder, Colorado, and topotypes of *uligocola* revealed no significant differences. These observations are indicative of 1) the differences between samples and populations which may be assigned to a single subspecies, and 2) the fact that in general these local differences are less than the differences between subspecies. From *insperatus*, the subspecies to the north, *uligocola* differs as follows: darker in both summer and winter pelage; averaging larger in most measurements of the skull; significantly longer molar tooth-row; hind foot averaging longer. For comparisons with the subspecies to the east and the northwest see the accounts of those below.

Remarks.—*M. p. uligocola* is more closely restricted to wet situations than *M. ochrogaster haydeni* (Baird) whose general range lies to the eastward. The numerous lakes, the continuous supply of water from the mountains, and the irrigation systems at lower altitudes along the eastern base of the mountains provide the conditions to which *uligocola* is suited. It is named for its predilection for water.

The variability in color is relatively greater in topotypes of both *uligocola* and of *modestus* than in many of the other series studied. Specimens from Denver and Colorado Springs taken in late autumn and winter (October to February) are paler, more reddish and less blackish, than specimens taken in June and July at Loveland. This reddishness results from longer, and more intensely reddish tips of the hair. The entire hairs also are longer. The average weight of 16 adults (12 males and 4 non-pregnant females) from near Loveland is 49.7 gms. The average length of the ear is 13.4.

Specimens examined.—Total 228. *Colorado*: LARIMER Co.: 6 mi. W, ½ mi. S Loveland, 5200 ft., 18; 3 mi. N Loveland, 3; Loveland, 4 (USBS). MORGAN Co.: 4 mi. W Orchard, 4 (Mich); Orchard, 1 (Mich). BOULDER Co.: Boulder, 91 (USNM 19, UC 12, Chi 60 examined, additional specimens in the collection); Valmont, 4 (UC). CLEAR CREEK Co.: Clear Creek, N side of Idaho Springs, 1. JEFFERSON Co.: Olivet, 10 (Denv). ADAMS Co.: Crook's Lake, 8 (Denv); Barr, 16 (Denv 15, ERW 1). ARAPAHOE Co.: Denver, 21 (USBS 3, AMNH 8, Denv 10). PARK Co.: William's Ranch, near Tarryall, 11 (Denv). TELLER Co.: Divide, 2 (ERW). EL PASO Co.: Colorado Springs, 31 (ERW 22, AMNH 7, MVZ 2); 12 mi. S Colorado Springs, 3 (MVZ).

***Microtus pennsylvanicus finitus* new subspecies**

Type.—Adult female, skin and skull, number 50204, University of Kansas, Museum of Natural History, obtained by J. K. Jones, Jr., original number 906, 5 miles north and 2 miles west of Parks in Dundy County, Nebraska, on August 16, 1952.

Range.—The valley of the north fork of the Republican River in eastern Colorado and southwestern Nebraska.

Diagnosis.—Entire animal and skull large; color dark for the species; zygomatic breadth large; upper molars large and upper molar tooth-row relatively long; braincase elongate; auditory meatus relatively small; bullae large; incisors relatively procumbent.

Comparisons.—From *M. p. uligocola*, the subspecies to the west, *M. p. finitus* differs in darker color. These subspecies resemble each other in large size and large molar teeth. A comparison of nine pairs of skulls by the "method of pairs" shows three features in which *finitus* (from at, or near, the type locality) differs from *uligocola* (from at, or near, the type locality); in 7 of 9 pairs *finitus* had a relatively more elongate braincase (Confidence Limit .60); in 9 of 9 pairs *uligocola* had larger auditory meatuses (C. L. .99); in 7 of 9 pairs *uligocola* had relatively larger bullae (C. L. .80). From *M. p. pennsylvanicus* (Ord) from eastern Nebraska and eastern South Dakota *finitus* differs in larger size; darker color; larger molar teeth and longer upper molar tooth-row. Five pairs of skulls were compared; in all 5 pairs, *finitus* had more procumbent incisor teeth (C. L. .97) and wider first upper molars (C. L. .97); and in 4 of 5 pairs *finitus* had a relatively more elongate braincase (C. L. .60). From *M. p. insperatus*, the subspecies to the north, *finitus* differs as follows: color darker, size larger, molar teeth relatively larger and alveolar length of the upper molar tooth-row greater. Five pairs of skulls were compared and in all 5 pairs *finitus* had more procumbent upper incisors (C. L. .97) than *insperatus*; in 4 of 5 pairs *insperatus* had a relatively more elongate braincase (C. L. .60), narrower first upper molariform tooth (C. L. .75), and shorter distance between the alveoli of the first upper molars (C. L. .88).

Remarks.—The species, *Microtus pennsylvanicus*, in Pleistocene time ranged onto the plains of Kansas as far southward as Meade County, Kansas (Hibbard, 1940:421). This occurrence indicates a cooler more humid climate then than now in southwestern Kansas. *M. p. finitus* is more closely associated with water than *Microtus ochrogaster*, the only other species of *Microtus* now occupying the same region, although both species have been captured at certain places in the same runways. In Nebraska, a marginal part of the range of the species, *M. pennsylvanicus* has been taken at scattered localities. This scattered and localized distribution of suitable habitat undoubtedly limits gene-flow between these relict populations. Presumably as a result of this isolation *finitus* has accumulated and maintained its distinctive characteristics. The subspecies is so named because of its limited range. The average weight of eight specimens (4 males and 4 non-pregnant females) from Dundy County is 57.2 grams.

Specimens examined.—Total 26. *Colorado*: YUMA Co.: Wray, 3 (USBS); 1 mi. W Laird, 2. *Nebraska*: DUNDY Co.: 5 mi. N, 2 mi. W Parks (Rock Creek State Fish Hatchery), 19; Haigler, 2 (USBS).

***Microtus pennsylvanicus pullatus* new subspecies**

Type.—Adult male, skin and skull, number 37873, University of Kansas, Museum of Natural History, obtained by Rollin H. Baker, original number 1343, 12 miles north and 2 miles east of Sage, 6100 ft., in Lincoln County, Wyoming, on July 19, 1950.

Range.—North-central Utah, eastern Idaho, western Wyoming, and southwestern Montana. See Figures 1 and 2.

Diagnosis.—Size average; color dark, especially in southern part of range; tail relatively long; molar teeth small; nasals relatively broad; maxillary septum relatively small.

Comparisons.—From *M. p. uligocola*, the subspecies to the southeast, *M. p. pullatus* differs as follows: relatively darker in southern part of its range (see Figure 1); smaller, tail relatively longer. In 6 of 7 pairs of skulls compared of *pullatus* (from Lincoln Co., Wyoming) and *uligocola* (from Larimer Co., Colorado), *pullatus* had relatively broader nasals (Confidence Limit .85); *uligocola* had larger maxillary septa (C. L. .97) and larger molar teeth (C. L. .90). From *insperatus*, the subspecies to the east, *pullatus* differs as follows: both summer and winter pelage darker; tail longer both actually and relatively; upper molar tooth-row shorter. Ten pairs of skulls of specimens from near Afton, Wyoming, representing *pullatus*, and from northeastern Wyoming, representing *insperatus*, revealed no significant differences in the features observed by the "method of pairs". Although not compared in detail with the subspecies to the north, *M. p. drummondi* (Audubon and Bachman), examination of specimens from western Montana and the accounts of other authors indicate that topotypes of *pullatus* are darker, longer-tailed, slightly larger-skulled and perhaps longer over all.

Remarks.—In this subspecies there is a cline in color from dark in extreme southwestern Wyoming to pale in north-central Wyoming and Montana as the range of *M. p. insperatus* is approached. There is thus a broad zone of intergradation in color and the line separating the subspecies must be drawn somewhat arbitrarily. In Wyoming the most distinct break in this cline is in the Big Horn Basin and if a detailed study of the species were made in Montana the break would probably be found where the mountains meet the plains, roughly as shown in Figure 1. There is a similar cline in western Montana in color. The mice are paler farther north as one approaches the Canadian border although they do not become so pale as *insperatus*. Darkness is a characteristic of several non-adjacent subspecies of *Microtus pennsylvanicus*, for example *M. p. kincaidi* Dalquest in central Washington (Dalquest, 1948:347), *M. p. finitus*, and *M. p. nigrans* Rhoads in eastern Virginia, but these subspecies presumably differ in other characters.

Some morphological features of the same kind and degree that differentiate subspecies in one place may not vary geographically in another place. Furthermore the geographic variation in one feature may be only partly correlated with the variation in another feature. The variation in *M. p. pullatus* is an example: Specimens from near Pocatello, Idaho, are darker than topotypes of *modestus* but specimens from Fremont County, Idaho, are indistinguishable from topotypes of *modestus* (Davis, 1939:315). I have examined a number of mice from the Bitterroot Valley in western Montana and the color value for 12 adults is 2.7. They are slightly but not significantly paler than topotypes of *modestus*. This is a result of the cline mentioned above and does not indicate relationship with *modestus*. Some average measurements of 10 skulls from this series are as follows: condylobasilar length, 24.9; zygomatic breadth, 14.4; interorbital breadth, 3.4; lambdoidal breadth, 11.5; prelambdoidal breadth, 9.1; alveolar length of upper molar teeth, 6.5; and depth of braincase, 7.6. Average external measurements of 9 specimens are as follows: total length, 157; length of tail, 36; length of hind feet, 19.4. Mice from the Bitterroot Valley were compared with topotypes of *modestus* by the "method of pairs," and *modestus* had a larger foramen magnum in 6 of 6 pairs (Confidence Limit .97) and larger first upper molar teeth in 5 of 6 pairs (C. L. .75).

A comparison of topotypes of *pullatus* with *modestus* shows a similar difference in the teeth, *modestus* being larger, but in the size of the foramen magnum there is no difference. A comparison of the measurements of *pullatus* (Bitterroot Valley), *pullatus* (near topotypes), and *modestus* (topotypes) shows that the two series of topotypes differ significantly in condylobasilar length (of borderline significance), zygomatic breadth and lambdoidal breadth (both of which vary greatly with age), and length of molar series; the specimens from the Bitterroot Valley agree with *pullatus* rather than *modestus* in all of these characters. The specimens from the Bitterroot Valley are smaller than *pullatus* (topotypes) in total length; they more closely resemble *modestus* in length of tail, and the hind foot is shorter than in either *modestus* or *pullatus* (which do not differ significantly). Specimens from western Montana resemble *modestus* in certain respects but in most respects resemble topotypes of *pullatus* and are referred to *pullatus*. Specimens from Coeur d'Alene, Idaho, and from Blackfoot, Montana (marginal records of *modestus*, Hall and Cockrum, 1953:410) may be referred to *drummondi*. Marginal records of *pullatus* in Montana to my knowledge are: Florence, Ravalli Co.; Highwood Mtns., Chouteau Co.; 7 mi. NE Hilger, Fergus Co.; 10 mi. NW Park City, Stillwater Co. (all represented by specimens in the USBS). The line separating *pullatus* and *drummondi* is tentatively drawn as shown in Figure 1.

Specimens examined.—Total 256. *Wyoming*: YELLOWSTONE NATIONAL PARK: Mammoth Hot Springs, 17 (USNM). PARK CO.: 3½ mi. E, ¾ mi. S Cody, 5020 ft., 15. TETON CO.: *Whetstone Creek*, 4 (Mich); 5 mi. N Moran, 13 (Mich); *Moran and environs (4 localities within 4 miles of Moran)*, 6200 ft., 54; *Teton Park*, Trappers Lake, 3 (Mich); *Teton Park*, *Jenny Lake*, 1 (Mich); *Teton Park*, *String Lake*, 1 (Mich); *Sheep Creek*, 1 (Mich); *Jackson and environs*, 115 (Mich 113, USBS 1). SUBLETTE CO.: 34 mi. N, 4 mi. W Pinedale, 7950 ft., 2; Kendall, 5 (Mich). LINCOLN CO.: 9½ mi. N, 2 mi. W Afton, 6100 ft., 1; 9 mi. N, 2 mi. W Afton, 3; 7 mi. N, 1 mi. W

Afton, 11; 15 mi. N, 3 mi. E Sage, 6100 ft., 1; 12 mi. N, 2 mi. E Sage, 4; 6 mi. N, 2 mi. E Sage, 2.

Microtus pennsylvanicus insperatus (Allen)

Arvicola insperatus Allen, Bull. Amer. Mus. Nat. Hist., 6:347, December 7, 1894.

Microtus pennsylvanicus insperatus, Anderson, Canadian Field-Nat., 57:92, October 17, 1943.

Microtus pennsylvanicus wahema Bailey, Jour. Mamm., 1:72, March 2, 1920.

Type.—Adult male, skin and skull, number 8105/6731 American Museum of Natural History, obtained by W. W. Granger, at Custer, Black Hills, South Dakota, August 9, 1894.

Range.—Western South Dakota, southwestern North Dakota, eastern Montana, southern Alberta and Saskatchewan, eastern Montana, and northeastern Wyoming.

Comparisons.—*Microtus pennsylvanicus insperatus* is paler than any adjacent subspecies. It has been compared with *pullatus* in the preceding account. Bailey's studies of *M. p. wahema* [= *insperatus*] and his comparisons with *M. p. pennsylvanicus* and *M. p. drummondi* to the east and north, in North Dakota (1920, 1927), are the basis for the northern and eastern boundaries of the range of *insperatus* in Figure 1. Comparison of *insperatus* with *finitus* to the south is made in the account of the latter.

**TABLE 1. AVERAGE MEASUREMENTS (IN MILLIMETERS) OF ADULT
MICROTUS PENNSYLVANICUS**

Locality (or area)	No. of adults averaged	Total length	Length of tail	Length of hind feet	Condylbasilar length, kull	Zygomatic breadth
<i>M. p. insperatus</i> , Wyoming						
Crook and Campbell Cos.	12	166.5	41.4	20.7	25.85	14.80
Sheridan County	20	169.5	46.7	21.2	26.20	15.25
Johnson County	20	167.8	47.9	20.7	25.67	15.29
Weston County, mean	15	161.0	39.8	20.7	25.55	14.91
" Co., stand. dev.	...	11.3	3.0	0.6	.82	.43
" Co., minimum	...	150.	35.	20.	24.2	14.2

Crook and Campbell Cos.	12	11.70	9.05	6.90	3.52	7.92
Sheridan County	20	12.20	9.33	6.90	3.55	7.95
Johnson County	20	12.04	9.02	7.14	3.52	7.84
Weston Co., mean	15	12.03	9.25	6.84	3.60	8.02
" "						
stand. dev.	...	.40	.37	.16	.13	.23
" "						
minimum	...	11.4	8.7	6.6	3.4	7.7
" "						
maximum	...	12.7	9.9	7.2	3.8	8.5
<i>M. p. pullatus</i> , Wyoming						
Park County	10	12.34	9.52	6.72	3.50	8.30
Teton County	20	11.58	8.98	6.77	3.54	7.66
Sage, Lincoln Co.	6	11.87	9.08	6.77	3.58	7.88
Afton, " "						
mean	14	11.74	9.06	6.53	3.55	7.90
" , stand.						
dev.	...	.50	.15	.23	.12	.43
" ,						
minimum	...	11.0	8.9	6.1	3.2	7.1
" ,						
maximum	...	12.5	9.4	7.0	3.6	8.2
<i>M. p. uligocola</i> , Colorado						
Park Co. (Denv)	7	12.20	9.40	7.07	3.60	7.91
Boulder Co. (Chi)	30	12.14	9.24	7.07	3.36	7.85
Denver	8	12.23	9.35	7.27	3.55	7.99
Colorado Springs	16	12.18	9.22	7.20	3.43	7.89
Loveland, mean	16	12.31	9.46	7.14	3.59	8.14
" , stand.						
dev.	...	.65	.41	.32	.16	.42
" ,						
minimum	...	11.4	9.0	6.8	3.3	7.5

" , maximum	...	13.8	10.6	7.6	3.8	9.4
<i>M. p. finitus</i>						
Wray, Colorado	3	12.30	9.43	7.40	3.53	8.23
Dundy Co., Nebr., mean	12	12.64	9.39	7.52	3.66	8.22
" " , stand. dev.	...	.61	.40	.31	.26	.43
" " , minimum	...	11.6	8.5	7.0	3.2	7.6
" " , maximum	...	13.5	9.8	7.9	3.9	8.7
<i>M. p. modestus</i> , Colorado						
Alamosa, Colorado	3	11.97	9.27	6.83	3.60	8.01
Cochetopa Pass, Colo.	25 ¹	12.16	9.04	6.81	3.54	8.18
" , stand. dev.	...	.37	.31	.21	.14	.23
" , minimum	...	11.4	8.6	6.5	3.3	7.7
" , maximum	...	13.1	10.0	7.5	3.9	8.8

¹ 29 specimens were used in taking measurements of the skull.

Remarks.—Bailey (1900:20) had only 7 specimens from northeastern Wyoming and western South Dakota, of *M. pennsylvanicus* and thought that *Arvicola insperatus* Allen (1894:347) was not subspecifically distinct from *modestus*. Subsequently Bailey (1920:72) had adequate numbers of specimens and described *Microtus pennsylvanicus wahema* from eastern Montana and western North Dakota. Anderson (1943:92) concluded that *wahema* was not distinct from *insperatus* and therefore the name *M. p. insperatus* (Allen) is applicable to this subspecies. On the basis of specimens that I have examined from Montana, South Dakota, and Wyoming I concur with Anderson. Bailey's characterization of *wahema* is applicable to *insperatus* as I interpret it. In the Black Hills, *Microtus longicaudus longicaudus* (Merriam) occurs together with *insperatus*.

Specimens examined.—Total 123. *Wyoming*: SHERIDAN CO.: 3 mi. WNW Monarch (=Kleeburn), 3800 ft., 4; 4 mi. NNE Banner, 4100 ft., 26; 5 mi. NE Clearmont, 3900 ft., 3. JOHNSON CO.: 5½ mi. W, 1 mi. S Buffalo, 5520 ft., 1; 5½ mi. W, 1½ mi. S Buffalo, 1; 1 mi. W, 4/5 mi. S Buffalo, 4800 ft., 36; ¼ mi. E Klondike, 5160 ft., 1. CAMPBELL CO.: Belle Fourche River,

45 mi. S, 13 mi. W Gillette, 5350 ft., 2. CROOK Co.: 3 mi. S, 2 mi. E Rocky Point, 3800 ft., 6; Bear Lodge Mts., 6½ mi. SSE Alva, 1 (Mich); 15 mi. N Sundance, 5500 ft., 3; 15 mi. ENE Sundance, 3825 ft., 6; 3 mi. NW Sundance, 5900 ft., 1; 1⅓ mi. NW Sundance, 5000 ft., 4; Sundance, 1 (USBS). WESTON Co.: 1½ mi. E Buckhorn, 6150 ft., 26; Newcastle, 1 (USBS).

GENERAL REMARKS

The region considered in this paper differs in several regards from the state of Pennsylvania, where variation in the skulls of this species has been studied in detail by Snyder (1954) who referred all populations there to a single subspecies. In some characteristics of the skulls, populations within Pennsylvania differed as much or more than the subspecies from Wyoming and Colorado. In other characteristics of the skulls and of the skins differences are greater between populations in Wyoming and Colorado. The region discussed here is approximately five times as large as the state of Pennsylvania. Populations of *M. pennsylvanicus* are less continuously distributed than in Pennsylvania owing to major physiographic and climatic barriers and also owing to competition with one or more of the five other species of *Microtus* occurring in this region. The distribution of three of these species has been discussed by Findley (1945:419). Large areas of relatively greater aridity, such as the region occupied by the subspecies *insperatus*, occur in Wyoming and Colorado. I have pointed out that the populations which I have designated as subspecies are not absolutely uniform. Also the different subspecies are not of exactly equal degrees of difference. However, there is considerable uniformity of populations occupying conveniently mapped geographic areas. In my opinion, the use of subspecific nomenclature is justified in this case, although not completely unambiguous.

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